

NO ARBITRAGE IN INSURANCE AND THE QP-RULE

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ABSTRACT. This paper is an attempt to study fundamentally the valuation of insurance contracts. We start from the observation that insurance contracts are inherently linked to financial markets, be it via interest rates, or – as in hybrid products, equity-linked life insurance and variable annuities – directly to stocks or indices. By defining portfolio strategies on an insurance portfolio and combining them with financial trading strategies we arrive at the notion of insurance-finance arbitrage (IFA). A fundamental theorem provides two sufficient conditions for presence or absence of IFA, respectively. For the first one it utilizes the conditional law of large numbers and risk-neutral valuation. As a key result we obtain a simple valuation rule, called QP-rule, which is market consistent and excludes IFA.

Utilizing the theory of enlargements of filtrations we construct a tractable framework for general valuation results, working under weak assumptions. The generality of the approach allows to incorporate many important aspects, like mortality risk or dependence of mortality and stock markets which is of utmost importance in the recent corona crisis. For practical applications, we provide an affine formulation which leads to explicit valuation formulas for a large class of hybrid products.

Keywords: arbitrage, fundamental theorem of asset pricing, insurance valuation, variable annuities, hybrid products, equity-linked life insurance, enlargement of filtration, affine processes, longevity risk, corona crisis, QP-rule.

1. INTRODUCTION

This paper is a first step into the direction of defining and finding arbitrage-free premiums of insurance products. We start from the observation that insurance contracts are inherently linked to financial markets, be it via interest rates, or via direct links of the contractual benefits to stocks or indices. Insurance companies hence naturally form a portfolio of insurance contracts for pooling risks and combine them with trading strategies on financial markets. We relate insurance arbitrage to the size of premiums and possible insurance-finance arbitrages.

1.1. Effect of institutional differences. If it is true that a share, a call option and an insurance contract all qualify as *securities* defined as *the legal representation of the right to receive prospective future benefits under stated conditions*, see [Sharpe et al. \(1999\)](#), the environment of current trade varies with the security to be sold or bought. While the *share* is anonymous in terms of buyer and seller, a *call* carries the information of the party who wrote it, an *insurance contract* can be sold only by an office and to a person who explicitly appears in the contract, see for example [Boudreault and Renaud \(2019\)](#). In contrast to financial markets, participants in primary insurance markets are a priori separated into two groups: potential sellers (insurance companies) who are to provide *contracts* contingent to insurance events and potential buyers (insurance seekers) who will provide premiums as current cash. Secondary markets for insurance are rare and therefore not considered here.

1.2. Necessity of a definition of insurance arbitrage. The definition of (financial) arbitrage as *the simultaneous purchase and sale of the same, or essentially similar, security in two different markets for advantageously different prices*, see [Sharpe et al. \(1999\)](#), relies on *despite whatever finance related event happens*-type statements and may require the insurance seeker to act as a seller of contracts, a rather rarely allowed case. Hence we have to find an appropriate definition of insurance arbitrage as a (first) step towards valuation. It is natural to try to express arbitraging insurance payments (cash in-flows and out-flows) by looking at the *with probability one insurance related events*

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resulting from the strong law of large numbers. Notice that we face an asymmetry between seller(s) and buyers of insurance, as insurance contracts are securities which are not liquidly traded receivables. To deal with them we first construct a particular linear measure, which values financial securities by risk-neutral pricing while using the private view of insurance claims.

From a more fundamental viewpoint, we aim at judging a candidate premium p for a considered insurance contract. To this end, we first define absence of insurance arbitrage starting from p . Then, we introduce additional trading on a financial discrete-time market by relying on classical self-financed trading strategies and the combination leads to our notion of *insurance-finance arbitrage* (IFA). As a next step, we show that bounding p suitably from above guarantees the absence of insurance arbitrage (NIFA) which we find a reasonable condition to the status of valuation. To the best of our knowledge this requirement is new in the literature.

The mathematical foundation of our approach lies in considering two filtrations: the publicly available information and the (additional) private information of the insurance company. We therefore utilize the theory of enlargements of filtration to provide tractable environments for the valuation of general insurance claims. In an immersion setting we constitute a general affine framework which is highly tractable and allows for the valuation of insurance products related to financial markets excluding insurance-finance arbitrage.

Since our setting is very general, we are able to generalize over existing approaches: first, we consider a financial market without restricting to the complete case. Second, we allow for arbitrary dependence between the financial market and the insurance quantities - this is important in many insurance products, for example when considering surrender behavior or stochastic mortality. The recent corona outbreak highlights the necessity of allowing dependence between mortality and stock markets. Third, we provide a fundamental justification of our valuation rule by relying on the (conditional) strong law of large numbers on the insurance side and on risk-neutral pricing on the financial side. Finally, we provide a general affine framework for insurance products related to financial markets. Among many other effects, this allows to incorporate longevity risk via stochastic mortality rates.

1.3. Related literature. The valuation of insurance contracts is a problem which has been intensively studied in the literature, see for example [Wüthrich and Merz \(2013\)](#), [Asmussen and Albrecher \(2010\)](#) and references therein. The existing approaches to the valuation of insurance products linked to financial market can be divided into five classes.

First, it is common in practice to value directly *under some ad-hoc chosen risk-neutral measure*, see for example [Brennan and Schwartz \(1976\)](#), [Dai et al. \(2008\)](#), [Krayzler et al. \(2015\)](#), [Cui et al. \(2017\)](#) and references therein. This approach is quite popular, as it often leads to explicit results in a direct matter.

Second, the *benchmark approach* with its application to the problem at hand introduced in [Bühlmann and Platen \(2003\)](#) and its application to financial markets from [Platen \(2006\)](#). It uses the growth-optimal portfolio (GOP) as numéraire and evaluates products by taking expectations under P , see e.g. [Biagini et al. \(2015\)](#) for an insurance application.

Third, the *local risk-minimization approach*, specifying the risk-neutral measure on the insurance side with a risk-minimizing procedure, see for example [Møller \(2001\)](#), [Pansera \(2012\)](#). It is assumed in these approaches that insurance risks, like mortality risk, can be diversified away.

Fourth, *indifference pricing* leads to a non-linear pricing rule and we refer to [Blanchet-Scalliet et al. \(2015\)](#), [Chevalier et al. \(2016\)](#) for details and further literature. Finally, there are valuation methodologies utilizing risk measures, often based on an axiomatic view, see [Tsanakas and Desli \(2005\)](#), [Pelsser and Stadje \(2014\)](#), or on hedging, see [Chen et al. \(2020\)](#).

Some insurance products we have in mind are variable annuities, which explicitly link the insurance benefits to the performance of financial markets. We refer to [Bacinello et al. \(2011\)](#) for an extensive overview of related literature.

1.4. Structure of the paper. In Section 2 we introduce a linear valuation rule for insurance claims which we call QP-rule. This rule combines risk neutral pricing (under Q) with actuarial valuation (under P). In Section 3 we provide a fundamental theorem which relates NIFA on insurance-finance markets to a majorization of premiums. The QP-rule arises as a natural valuation tool excluding IFA, see in particular Remarks 3.8 and 3.10. In Section 4 we introduce the connection to enlargements

of filtration and in Section 5 we provide an affine framework. Section 6 discusses longevity risk and computes values explicitly for some examples. Section 7 concludes the paper.

2. VALUATION OF NON-TRADED WEALTH

For insurance companies, the valuation of non-traded wealth is a core topic: indeed, as mentioned in the introduction, insurance contracts are not traded. In addition, they have direct links to financial markets (interest rates, variable annuities, equity-linked life insurance etc.), hence a consistent valuation rule linking traded assets with non-traded insurance contracts is of highest importance.

In a first step, this leads to the following measure theoretic problem: given two σ -algebras $\mathcal{F} \subset \mathcal{G}$ on a set Ω , a probability Q on $(\Omega, \mathcal{F})$, and a probability P on $(\Omega, \mathcal{G})$, how should we extend Q to $(\Omega, \mathcal{G})$ in a P -reasonable way. We assume that Q and $P|_{\mathcal{F}}$ are equivalent; here $P|_{\mathcal{F}}$ denotes the restriction of P to $\mathcal{F}$. In Section 2.2 we extend this problem to the dynamic setting with two filtrations.

2.1. Extending a measure to a larger σ -algebra. The following proposition gives a unique extension of Q with P .

Proposition 2.1. *There is a unique probability measure, denoted by $Q \odot P$, on $(\Omega, \mathcal{G})$ such that $Q \odot P = Q$ on $\mathcal{F}$ and for all $G \in \mathcal{G}$ it holds that $Q \odot P(G|\mathcal{F}) = P(G|\mathcal{F})$. This measure $Q \odot P$ satisfies for any random variable $X \geq 0$ that*

(i) *for a σ -algebra $\mathcal{H} \subset \mathcal{F}$,*

$$E_{Q \odot P}[X|\mathcal{H}] = E_Q[E_P[X|\mathcal{F}]|\mathcal{H}],$$

(ii) *for a σ -algebra $\mathcal{H}$ satisfying $\mathcal{F} \subset \mathcal{H} \subset \mathcal{G}$,*

$$E_{Q \odot P}[X|\mathcal{H}] = E_P[X|\mathcal{H}].$$

Letting $\mathcal{H} = \{\emptyset, \Omega\}$, we obtain from (i) the one-period QP -rule:

$$E_{Q \odot P}[X] = E_Q[E_P[X|\mathcal{F}]]. \quad (1)$$

Proof. We construct the measure $Q \odot P$ as follows: let L denote the Radon-Nikodym derivative of Q with respect to $P|_{\mathcal{F}}$, such that $dQ = LdP$. Define $Q \odot P$ on $(\Omega, \mathcal{G})$ by

$$d(Q \odot P) = LdP. \quad (2)$$

such that $Q \odot P(G) = E_P[\mathbb{1}_G L]$. Obviously $Q \odot P$ and Q coincide on $\mathcal{F}$ by construction. Moreover, for $G \in \mathcal{G}$ we have that

$$\begin{aligned} \int_F \mathbb{1}_G d(Q \odot P) &= \int_F \mathbb{1}_G L dP = \int_F LP(G|\mathcal{F}) dP \\ &= \int_F P(G|\mathcal{F}) d(Q \odot P), \quad F \in \mathcal{F}, \end{aligned} \quad (3)$$

such that $Q \odot P(G|\mathcal{F}) = P(G|\mathcal{F})$. In view of (i), we obtain with $H \in \mathcal{H} \subset \mathcal{F}$, that $H \in \mathcal{F}$, and it also follows that $Q \odot P(G|\mathcal{H}) = P(G|\mathcal{H})$. Moreover, letting $H = \Omega$ we obtain

$$\int \mathbb{1}_G d(Q \odot P) = \int LP(G|\mathcal{H}) dP = \int P(G|\mathcal{H}) dQ \quad (4)$$

and an application of the monotone class theorem yields (i).

For uniqueness, consider a further measure R , such that $R = Q$ on $\mathcal{F}$ and such that $R(G|\mathcal{F}) = P(G|\mathcal{F})$. Then, for $G \in \mathcal{G}$ it holds that

$$R(G) = \int R(G|\mathcal{F}) dR = \int P(G|\mathcal{F}) dR = \int P(G|\mathcal{F}) dQ = Q \odot P(G) \quad (5)$$

and we obtain that $Q \odot P = R$.

Finally, in view of (ii) we observe that by Bayes' rule, it follows from $\mathcal{F} \subset \mathcal{H} \subset \mathcal{G}$ that

$$E_{Q \odot P}[X|\mathcal{H}] = \frac{E_P[L_T X | \mathcal{H}]}{E_P[L_T | \mathcal{H}]} = E_P[X|\mathcal{H}],$$

since L_T is $\mathcal{F}$ -measurable and hence $\mathcal{H}$ -measurable. $\square$

The idea of choosing the Radon-Nikodym derivative on the smaller σ -algebra as density on the larger probability space may already be found in Plachky and Rüschendorf (1984) who are studying measure extensions in a statistical context. In the financial context, already Dybvig (1992) studied this extension in the framework of state-price densities.

For the application we have in mind, the valuation of a non-traded insurance contract affected by financial markets has to be solved. In this case, the σ -algebra $\mathcal{F}$ consists of publicly available information on financial markets, and in particular on the traded assets. The larger σ -algebra $\mathcal{G}$ contains private information available to the insurance company and the non-traded insurance liability is considered as a random variable on the larger probability space $(\Omega, \mathcal{G})$.

We assume that the financial market $(\mathcal{F}, S)$ is free of arbitrage with respect to publicly available information $\mathcal{F}$ (in particular trading strategies adapted to $\mathcal{F}$ are being used). This is equivalent to existence of a risk-neutral measure Q on $(\Omega, \mathcal{F})$. Hence, Proposition 2.1 applies and allows, in the spirit of Dybvig (1992), to value the untraded insurance liability X : equation (1), which we call *QP-rule*, gives a value of X which does exclude financial arbitrage. We will also show that the QP-rule is *market-consistent* in Section 2.3.

Example 2.2 (A discrete example). Consider a one-period model with three states on the financial market: good, medium, and bad (g, m, b) . We assume two states on the insurance side: payment and no payment $(p, \bar{p})$, corresponding to

$$\Omega = \{(g, p), (g, \bar{p}), (m, p), (m, \bar{p}), (b, p), (b, \bar{p})\}.$$

Denote the good, medium and bad states by $g = \{(g, p), (g, \bar{p})\}$, $m = \{(m, p), (m, \bar{p})\}$, and $b = \{(b, p), (b, \bar{p})\}$. The information on the financial market is given by $\mathcal{F} = \sigma(g, m, b)$, while $\mathcal{G} = \mathcal{P}(\Omega)$.

The risky asset S satisfies $S_0 = 1$ and $S_1(g) = 1.5$, $S_1(m) = 1$, and $S_1(b) = 0.5$. The numéraire is set constant and equal to 1.

We assume that the measure P is given by $(0.1h, 0.9h, 0.2k, 0.8k, 0.4l, 0.6l)$ with $h + k + l = 1$, $h, k, l \in [0, 1]$.

Complete market: if $k = 0$ and $h, l \in (0, 1)$, the financial market is binomial and therefore complete. The unique risk-neutral measure is given by $Q(g) = 1/2$.

For the application of (1) we compute the conditional probabilities

$$\begin{aligned} P(\cdot | g) &= (0.1, 0.9, 0, 0, 0, 0), \\ P(\cdot | b) &= (0, 0, 0, 0, 0.4, 0.6). \end{aligned} \tag{6}$$

The measure $Q \odot P$ is given as the convex combination

$$Q \odot P(A) = \mathbf{1}_{\{g\}} \cdot Q(g) \cdot P(A|g) + \mathbf{1}_{\{b\}} \cdot Q(b) \cdot P(A|b),$$

s.t. $Q \odot P$ equals $(0.05, 0.45, 0, 0, 0.2, 0.3)$, independently of the value of h and l .

Incomplete market: if $h, k, l \in (0, 1)$, the market is incomplete. The set of risk-neutral measures is given by $Q^s(g) = s$, $Q^s(m) = 1 - 2s$, and $Q^s(b) = s$, with $s \in (0, 1/2)$. In addition to (6) we have $P(\cdot | m) = (0, 0, 0.2, 0.8, 0, 0)$. We obtain

$$Q^s \odot P(A) = \mathbf{1}_{\{g\}} \cdot Q^s(g) \cdot P(A|g) + \mathbf{1}_{\{m\}} \cdot Q^s(m) \cdot P(A|m) + \mathbf{1}_{\{b\}} \cdot Q^s(b) \cdot P(A|b),$$

which gives $Q^s \odot P$ as $(0.1s, 0.9s, 0.2(1 - 2s), 0.8(1 - 2s), 0.4s, 0.6s)$. $\diamond$

2.2. The case of two filtrations. In this section we extend Proposition 2.1 to multiple periods. The time space $\mathbb{T} = \{0, 1, \dots, T\}$ is considered to be discrete and finite with $T < \infty$ ¹. We consider a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{T}}$ such that $\mathcal{F}_T \subset \mathcal{G}$.

We start with a stylized example which illustrates the most important feature of the insurance contracts we intend to value. In this regard, we call a stochastic process $(\tilde{A}_t)_{t \in \mathbb{T}}$ a *payment flow*, if $\tilde{A}_t \geq 0$ for all $t \in \mathbb{T}$. Let $(\tilde{S}_t^0)_{t \in \mathbb{T}} > 0$ be the numéraire process and denote by $A_t = \tilde{A}_t / \tilde{S}_t^0$, $t \in \mathbb{T}$ the associated discounted payment flow. Negative cash flows can be incorporated by considering positive and negative parts separately.

Example 2.3 (Stochastic mortality). Consider an insurance contract whose payments may be *separated* into two components: on the one side a random time τ and on the other side a discounted payment flow A ; the insurance offers the payment of $\tilde{A}_t$ at t if $\tau = t$. We assume that the payment flow A is $\mathbb{F}$ -adapted. The random time τ can be used to model the random life time in the case

¹All our results hold also for $T = \infty$ under the condition that there exists a martingale measure Q on $(\Omega, \mathcal{F}_\infty)$.

of a life insurance, or the time of surrender in a variable annuity or the time of a catastrophe in CAT-bonds.

Consider an $\mathbb{F}$ -adapted, increasing process $\Lambda = (\Lambda_t)_{t \in \mathbb{T}}$ with $\Lambda_0 = 0$. Assume that there exists a standard exponential random variable E which is independent of the financial market (i.e. independent of $\mathcal{F} = \mathcal{F}_T$) and let²

$$\tau = \inf\{t \in \mathbb{T} : \Lambda_t \geq E\}, \quad (7)$$

with the convention that $\inf \emptyset = T$. Then, τ is not an $\mathbb{F}$ -stopping time, but it is a stopping time in the progressive enlargement³ given by $\mathcal{G}_t = \mathcal{F}_t \vee (\tau \wedge t)$, $t \in \mathbb{T}$. Let $\mathcal{G} = \mathcal{F}_T \vee \tau$.

We are interested in the market-consistent valuation of the insurance contract. Its value at time 0 computes⁴ to

$$\sum_{t \in \mathbb{T}} E_{Q \odot P}[A_t \mathbb{1}_{\{\tau=t\}}]. \quad (8)$$

Using Proposition 2.1 and exploiting the independence between E and $\mathcal{F}$ we obtain that

$$Q \odot P(\tau > t | \mathcal{F}) = P(\tau > t | \mathcal{F}) = P(\Lambda_t < E | \mathcal{F}) = e^{-\Lambda_t} =: G_t, \quad (9)$$

and, hence, $P(\tau = t | \mathcal{F}) = G_{t-1} - G_t$. By (1) we obtain that

$$\begin{aligned} E_{Q \odot P}[A_\tau] &= \sum_{t \in \mathbb{T}} E_Q[A_t P(\tau = t | \mathcal{F})] \\ &= \sum_{t \in \mathbb{T}} E_Q[A_t (G_{t-1} - G_t)] \end{aligned}$$

and the remaining computations can be done as usual in a risk-neutral manner.

Stochastic mortality is achieved in this example by the (random) cumulated intensity process Λ . This is important for the modeling of longevity risk, see Section 6.3, or to incorporate dependencies between mortality and the stock market. The recent outbreak of the corona virus and the related infections had a dramatic impact on the stock market together with increased mortality; see Feng and Garrido (2011) for incorporating pandemic modeling into an actuarial context. The setting considered in this example allows to model this. $\diamond$

The example above has numerous applications in insurance: any insurance contract whose payoffs paid at t when $\tau = t$ and which are depending only on publicly available information (interest rates, stock markets, etc.). From Proposition 2.1 we obtain that for $0 \leq t \leq T$,

$$E_{Q \odot P}[X | \mathcal{F}_t] = E_Q[E_P[X | \mathcal{F}_T] | \mathcal{F}_t], \quad (10)$$

for any $X \in L^1$ or bounded from below.

While it might in general be difficult to compute the conditional expectations in Equation (10), under additional assumptions like immersion one is able to obtain tractable results. We exploit this in Section 4.

2.3. Market-consistent evaluation. In this section we briefly discuss the important property of market-consistency, see for example Cheridito et al. (2008), Artzner and Eisele (2010), Pelsler and Stadje (2014). Using equation (10) we extend (1) and obtain the *QP-rule* for the insurance contract introduced in Example 2.3: the value at time t of the discounted payment flow A on $(t, T]$ is

$$\Pi_t(A) := \sum_{s=t+1}^T E_{Q \odot P}[A_s \mathbb{1}_{\{\tau=s\}} | \mathcal{F}_t]. \quad (11)$$

The QP-rule ensures *market-consistent evaluation*, a necessary requirement under Solvency II: a *valuation rule* $\Pi' = (\Pi'_t)_{t \in \mathbb{T}}$ is a family of mappings with $\Pi'_t : L^0(\mathcal{G}) \rightarrow L^0(\mathcal{F}_t)$. Here $L^0(\mathcal{G})$ denotes the space of $\mathcal{G}$ -measurable random variables, and $L^0(\mathcal{F}_t)$ the space of $\mathcal{F}_t$ -measurable random variables, respectively. The valuation at time t , Π'_t , maps a discounted payoff H , i.e. a $\mathcal{G}$ -measurable random variable, to its value at time t , i.e. to an $\mathcal{F}_t$ -measurable random variable.

²Such a construction is called *doubly stochastic*. We refer to Bielecki and Rutkowski (2002), Brémaud (1981), Gehmlich and Schmidt (2018) for details and references. If $\Lambda_t = \int_0^t \lambda_s ds$, then λ is called the *intensity* of τ . If the intensity is deterministic, it is called *hazard rate*.

³For further details see Section 4.

⁴For the moment we simply apply Proposition 2.1, fundamental justifications will follow in Sections 2.3 and 3.

The valuation rule Π' is called *market-consistent* if the following holds: the value of a discounted financial payoff of the form $H + H'$ consisting of a financial part $H \in L^0(\mathcal{F}_T)$ and an actuarial part $H' \in L^0(\mathcal{G})$ with H and H' being bounded satisfies

$$\Pi'_t(H + H') = E_Q[H|\mathcal{F}_t] + \Pi'_t(H') \quad (12)$$

for some $Q \in \mathcal{M}_e(\mathbb{F})$. It follows immediately that the QP-rule and, in particular, the valuation rule Π given in (11), is market-consistent.

In addition to market consistency, the fundamental valuation theorem proved in the following section will show that the QP-rule excludes financial and insurance arbitrages. For the precise statement we refer to Theorem 3.6 and Remark 3.10.

The approach in Chen et al. (2020) introduces a concept of valuation, called *fair*, if it is market consistent and in addition coincided with valuation under P for payoffs which are independent of the future stock evolution (called *t-orthogonal*). The QP-rule satisfies also this property.

3. ARBITRAGE IN INSURANCE

In this section we start from a valuation rule p , like in Section 2.3, and study whether p is indeed fit for the job. For this we introduce the notion of *insurance arbitrage*. Insurance arbitrage is defined under the assumption that the insurer has the possibility to contract with a large number of clients so that he may reduce risk substantially. If this leads to a profit without the risk of losing money it will be called, in analogy to financial markets, an insurance arbitrage.

In addition we will allow for trading in a finite number of financial assets which will possibly lead to *financial arbitrage* and a combination of both approaches will be called *insurance-finance arbitrage* (IFA).

We use the QP-rule of Section 2, and prove that certain inequalities between p and the QP-value ensure the absence of insurance-finance arbitrage.

3.1. Definition of the insurance-finance arbitrage. In this paper we consider only the case of a single insurer. We assume that he can contract with possibly infinitely many *insurance seekers*. Insurance contracts are always of the form that a future claim is covered in exchange of a single premium paid at initiation of the insurance contract.

The insurance company has additional information (e.g. survival times or health state) which we describe by the filtration $\mathbb{G}$. Since the insurance company has also access to the publicly available information $\mathbb{F}$, we assume that $\mathcal{F}_t \subset \mathcal{G}_t$ for all $t \in \mathbb{T}$.

3.1.1. The insurance contracts. At each time $t \in \{0, 1, \dots, T-1\}$, the insurer has the possibility to issue contracts of the same type (depending on t). Each *insurance contract* exchanges (future) benefits against a premium paid at initiation⁵. Without loss of generality, we assume that all claims are settled at the maturity date T . The *benefits* of the insurance contract issued at time t are described by a $\mathcal{G}_T$ -measurable non-negative random variable $X_{t,T}$ (already discounted). This could be claims arising from life insurance in the time interval $(t, T]$ or from a variable annuity. The family $X = (X_{t,T})_{0 \leq t < T}$ is called the *family of benefits*.

The *premium* of the insurance contract $X_{t,T}$ issued at time t is denoted by p_t (already discounted); it is $\mathcal{G}_t$ -measurable. We set $p = (p_t)_{0 \leq t < T}$. The value p_t has to be regarded as a *basic part of premium* at time t for the risk $X_{t,T}$. On top, the insurance company adds a *commercial part* which contains additional costs, risk margins, etc. Risk margins are enforced by regulation and are non-linear rules: these topics will be treated with in a forthcoming paper.

To minimize risk on the insurance side, the insurer will exploit the possibility to form a portfolio of insurance contracts out of a population of *insurance seekers*. In this regard, we assume that at each time t the insurance can contract with possibly infinitely many insurance seekers. The associated $\mathcal{G}_T$ -measurable benefits are denoted by $X_{t,T}^1, X_{t,T}^2, \dots$.

It will turn out that the σ -algebra $\mathcal{G}_{t,T} = \mathcal{G}_t \vee \mathcal{F}_T$ containing the insurance information up to time t and all the publicly available information $\mathcal{F}_T$ plays a distinctive role. We make the following assumptions which generalize the classical framework of actuarial mathematics to stochastic assets and are supposed to correspond to the actual structure of the concrete commitments agreed upon by sellers and buyers of contracts:

⁵See section 3.3 on how to deal with scheduled premiums

Assumption 3.1. For each $t \in \mathbb{T}$,

- (i) $X_{t,T}^1, X_{t,T}^2, \dots$ are $\mathcal{G}_{t,T}$ -conditionally independent,
- (ii) $E[X_{t,T}^i | \mathcal{G}_{t,T}] = E[X_{t,T} | \mathcal{G}_{t,T}]$, $i = 1, 2, \dots$, and
- (iii) $\text{Var}(X_{t,T}^i | \mathcal{G}_{t,T}) = \text{Var}(X_{t,T} | \mathcal{G}_{t,T}) < \infty$, $i = 1, 2, \dots$

Inspired by large financial markets (see e.g. Klein (2000), Kabanov and Kramkov (1995), Klein et al. (2016)), we model the insurance portfolio as a limit of *allocations* of contracts with finitely many insurance seekers: an *allocation* $\psi_t = (\psi_t^i)_{i \geq 1}$ for time t is a finite-dimensional $\mathcal{G}_t$ -measurable, non-negative random variable, i.e. there exists an m such that $\psi_t^i = 0$ for $i \geq m$. Here, $\psi_t^i \in \mathbb{R}_+$ denotes the size of the contract with the i^{th} insurance seeker. In contrast to a financial trading strategy there is no rebalancing: if the insurance chooses an allocation at time t , then it is entailed to pay to the time- t -customers the associated claims at maturity.

This, together with the assumptions of homogeneity of premiums and benefits, induces the following notions: while the *commitment* of the insurer with allocation ψ_t at date t is given by the accumulated benefits, i.e.

$$\sum_{i \geq 1} \psi_t^i X_{t,T}^i,$$

its associated *value*, given by the accumulated premiums, equals

$$\sum_{i \geq 1} \psi_t^i p_t.$$

The profits and losses (P&L) of the allocation sequence $\psi = (\psi_t)_{t < T}$ amounts to the accumulated premiums minus the accumulated benefits. Summing up over all time instances $t = 0, \dots, T-1$ gives the P&L of the allocation ψ , which we denote by

$$V_T^I(\psi) := \sum_{t=0}^{T-1} \sum_{i \geq 1} \psi_t^i (p_t - X_{t,T}^i). \quad (13)$$

Finally, an *insurance portfolio strategy* ψ is a sequence $(\psi^n)_{n \geq 1}$ of (finite) allocations. We impose the following *admissibility conditions*:

- (i) *uniform boundedness*: there exists $C > 0$, such that

$$\|\psi_t^n\| := \sum_{i \geq 1} \psi_t^{n,i} \leq C \quad (14)$$

for all $n \geq 1$ and $0 \leq t < T$,

- (ii) *convergence of the total mass*: there exists $0 < \gamma_t \in \mathcal{F}_t$, such that

$$\|\psi_t^n\| \rightarrow \gamma_t \quad \text{a.s. for all } t < T, \quad (15)$$

- (iii) *convergence of the total wealth*: there exists a real-valued random variable $V = V^\psi$, such that

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) = V,$$

P -almost surely.

Let us remark that condition (14) implies uniform integrability for the sum of the $\mathcal{G}_{t,T}$ -conditional losses: indeed, if (A_m) is a decreasing sequence of $\mathcal{G}_{t,T}$ -measurable sets with $A_m \searrow \emptyset$, then

$$\begin{aligned} E\left[\mathbb{1}_{A_m} \sum_{t < T} \sum_{i \geq 1} \psi_t^{n,i} E[X_{t,T}^i | \mathcal{G}_{t,T}]\right] &= \sum_{t < T} E\left[\sum_{i \geq 1} \psi_t^{n,i} E[\mathbb{1}_{A_m} X_{t,T}^i | \mathcal{G}_{t,T}]\right] \\ &\leq C \sum_{t < T} E[\mathbb{1}_{A_m} X_{t,T}] \rightarrow 0 \end{aligned} \quad (16)$$

as $m \rightarrow \infty$, uniformly in n .

The assumption on $\gamma_t \in \mathcal{F}_t$ is weak: indeed, we often will use a total mass of $\gamma_t = 1 \in \mathcal{F}_t$ and consider allocations satisfying $\|\psi_t^n\| = 1$. These allocations distribute 1 in a $\mathcal{G}_t$ -measurable way between the insurance seekers. The assumption on convergence could be replaced by weaker ones, see Remark 3.4.

3.1.2. The financial market. The financial market consists of $d+1$ tradeable securities with $\mathbb{F}$ -adapted price process $\tilde{S} = (\tilde{S}^0, \tilde{S}^1, \dots, \tilde{S}^d)$. Traded assets could be bonds with and without credit risk, stocks, indices, etc. The information $\mathbb{F}$ will typically be strictly larger than the filtration generated by the traded assets: economic variables like employment rates or national mortality rates are examples of such publicly available information.

The numéraire $\tilde{S}^0$ with $\tilde{S}_0^0 = 1$ may be random, but is assumed to be strictly positive. Discounted price processes are denoted by $S = (S^1, \dots, S^d)$ where $S^i = \tilde{S}^i/\tilde{S}^0$, $i = 1, \dots, d$, and $S^0 \equiv 1$.

A trading strategy on the financial market is a d -dimensional, $\mathbb{F}$ -adapted process $\xi = (\xi_t)_{0 \leq t \leq T-1}$ with $\xi_t = (\xi_t^1, \dots, \xi_t^d)$. For a self-financing trading strategy ξ (see for example Föllmer and Schied (2011), Proposition 5.7), its P&L until T is given by

$$V_T^F(\xi) := \sum_{t=1}^{T-1} \xi_t \cdot \Delta S_t = \sum_{t=1}^{T-1} \sum_{i=1}^d \xi_t^i \cdot \Delta S_t^i,$$

with $\Delta S_t = S_{t+1} - S_t$. We assume that the market does not allow arbitrage so that there exist at least one equivalent martingale (or market-consistent) measure, i.e.

$$\mathcal{M}_e(\mathbb{F}) \neq \emptyset. \quad (17)$$

We emphasize that we do not assume completeness of the financial market $(\mathbb{F}, S)$ and that we allow for (financial) arbitrages when trading is done with the filtration $\mathbb{G}$. However, we assume that trading of the insurance company on the financial market is only done with respect to the public information in $\mathbb{F}$, thus coping with the law against *insider-trading*. On the other side, the insurance company of course uses the information $\mathbb{G}$ for the valuation and the allocation on the insurance contracts.

Summarizing, the *insurance-finance market* (X, p, S) consist of the three ingredients: the benefits X , the premiums p and the financial securities prices S .

Definition 3.2. On (X, p, S) we have an *insurance-finance arbitrage*, if there exist an admissible insurance portfolio strategy $(\psi^n)_{n \geq 1}$, and a self-financing trading strategy ξ such that

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \in L_0^+ \setminus \{0\}. \quad (18)$$

Otherwise, we say that there does not exist an *insurance-finance arbitrage* (NIFA).

It seems to be worthwhile pointing out again the asymmetry between the non-negative $\mathbb{G}$ -adapted processes ψ^n and the $\mathbb{F}$ -predictable strategy ξ without sign restrictions.

Remark 3.3 (On the role of the information). Depending on the application we will exploit the flexibility of choosing $\mathbb{F}$ in a suitable manner. In particular, it may be useful to assume that $\mathbb{F}$ contains also insurance information, like life tables or general factors which drive the evolution.

Remark 3.4 (On the convergence assumption). It may be noted that the existence of the limes in (18) is guaranteed by the third admissibility condition, convergence of the total wealth. This assumption could be weakened by replacing the limit in (18) by a limes inferior and then applying our results to the insurance portfolio strategy converging to this lower limit.

At the current stage, we provide a simple example which may serve as basis building block for a variable annuity. The valuation of a variable annuity will be described in detail in Section 6.

Example 3.5 (A simple example). Consider $d = 1$ and denote by $S = S^1$ the price of the stock. Let τ denote the lifetime of the insured person and consider the benefits $X_{t,T} = \mathbb{1}_{\{\tau > T\}} S_T$, i.e. a contract promising S_T given survival. Now if we have conditional independent copies of $\mathbb{1}_{\{\tau > T\}}$ with identical mean, assumed to satisfy $P(\tau > T | \mathcal{F}_T) = p(S_T)$, averaging by the strong law of large numbers (to be made precise below) leads to the $\mathcal{F}_T$ -conditional expectation

$$p(S_T) S_T,$$

which serves as a basis for valuation of this claim. Note that $p(S_T) S_T$ is $\mathcal{F}_T$ -measurable and hence turns out to be a European contingent claim which can be valued by standard risk-neutral valuation.

In this regard, for valuing this product we would consider $\mathbb{F}$ to be generated by S while the insurance information $\mathbb{G}$ is the progressive enlargement of $\mathbb{F}$ with τ . $\diamond$

3.2. Theorem on the absence of insurance-finance arbitrage. We begin by introducing certain upper bounds. Since the premium p_t is $\mathcal{G}_t$ -measurable, already the knowledge of p_t for an insurance seeker might offer arbitrage possibilities for her on the financial market. To avoid this, we introduce the $\mathcal{F}_t$ -conditional essential supremum (infimum) of p_t : for a family Ξ of random variables we define⁶

$$\text{ess sup}_{\mathcal{F}} \Xi := \text{ess inf} \{q \mid q \text{ is } \mathcal{F}\text{-measurable and } q \geq \xi \text{ for all } \xi \in \Xi\}. \quad (19)$$

Furthermore, denote

$$p_t^\uparrow = \text{ess sup}_{\mathcal{F}_t} p_t, \text{ and } p_t^\downarrow = \text{ess inf}_{\mathcal{F}_t} p_t. \quad (20)$$

Our main goal is the following fundamental theorem about the absence of insurance-finance arbitrage.

Theorem 3.6. *Let the insurance-finance market be described by (X, p, S) and assume that Assumption 3.1 holds.*

(i) *If there exists $Q \in \mathcal{M}_e(\mathbb{F})$ such that for all $t < T$*

$$p_t \leq E_{Q \otimes P}[X_{t,T} | \mathcal{F}_t], \quad P\text{-a.s.}, \quad (21)$$

then there is no insurance-finance arbitrage (NIFA).

(ii) *If there exists $t < T$, such that*

$$P\left(\bigcap_{Q \in \mathcal{M}_e(\mathbb{F})} \{p_t^\downarrow > E_{Q \otimes P}[X_{t,T} | \mathcal{F}_t]\}\right) > 0 \quad (22)$$

then there exists an insurance-finance arbitrage (IFA).

For the proof of Theorem 3.6 we need some preliminary results.

Proposition 3.7. *Under the Assumption 3.1 and the admissibility conditions (14) and (15) we have for all admissible allocation ψ and all $Q \in \mathcal{M}_e(\mathbb{F})$ that*

$$E_{Q \otimes P}\left[\lim_{n \rightarrow \infty} \sum_{i \geq 1} \psi_t^{n,i} p_t\right] = E_{Q \otimes P}[\gamma_t p_t], \quad \text{for all } t < T \quad \text{and} \quad (23)$$

$$E_{Q \otimes P}\left[\lim_{n \rightarrow \infty} \sum_{t=0}^{T-1} \sum_{i \geq 1} \psi_t^{n,i} X_{t,T}^i\right] = \sum_{t < T} E_{Q \otimes P}[\gamma_t X_{t,T}]. \quad (24)$$

Proof. Indeed, (23) follows from the convergence of the total mass in (15) together with dominated convergence. Recall that $V_T^I(\psi^n) = \sum_{t < T} \sum_{i \geq 1} \psi_t^{n,i} X_{t,T}^i$. As in Proposition 2.1, we denote the Radon-Nikodym density of Q with respect to $P|_{\mathcal{F}_T}$ by L_T . Then,

$$\begin{aligned} E_{Q \otimes P}\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n)\right] &= E_P\left[L_T \lim_{n \rightarrow \infty} V_T^I(\psi^n)\right] \\ &= E_P\left[L_T E_P\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n) | \mathcal{G}_{t,T}\right]\right]. \end{aligned}$$

For any $G \in \mathcal{G}_{t,T}$, we have

$$\begin{aligned} \int_G E_P\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n) | \mathcal{G}_{t,T}\right] dP &= \int_G \lim_{n \rightarrow \infty} V_T^I(\psi^n) dP \\ &= \int \liminf_{n \rightarrow \infty} \mathbf{1}_G V_T^I(\psi^n) dP \\ &\leq \liminf_{n \rightarrow \infty} \int_G V_T^I(\psi^n) dP = \liminf_{n \rightarrow \infty} \int_G E_P[V_T^I(\psi^n) | \mathcal{G}_{t,T}] dP. \end{aligned}$$

Furthermore,

$$\begin{aligned} E_P[V_T^I(\psi^n) | \mathcal{G}_{t,T}] &= \sum_{t < T} \sum_{i \geq 1} \psi_t^{n,i} E_P[X_{t,T}^i | \mathcal{G}_{t,T}] = \sum_{t < T} \sum_{i \geq 1} \psi_t^{n,i} E_P[X_{t,T} | \mathcal{G}_{t,T}] \\ &= \sum_{t < T} E_P[X_{t,T} | \mathcal{G}_{t,T}] \parallel \psi_t^n \parallel. \end{aligned}$$

⁶Compare Proposition 2.6 in Barron et al. (2003). In the following application Ξ will consist of only one random variable; we however take the freedom to define the conditional essential supremum for the more general case.

Hence,

$$\begin{aligned} \liminf_{n \rightarrow \infty} \int_G E_P[V_T^I(\psi^n) | \mathcal{G}_{t,T}] dP &= \liminf_{n \rightarrow \infty} \int_G \sum_{t < T} E_P[X_{t,T} | \mathcal{G}_{t,T}] \|\psi_t^n\| dP \\ &= \int_G E_P\left[\sum_{t < T} \gamma_t X_{t,T} | \mathcal{G}_{t,T}\right] dP, \end{aligned}$$

where we used uniform boundedness, (15), and dominated convergence for the last equality. We obtain that

$$E_{Q \odot P}\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n)\right] \leq \sum_{t < T} E_{Q \odot P}[\gamma_t X_{t,T}]. \quad (25)$$

On the other side, for any $G \in \mathcal{G}_{t,T}$, we have

$$\begin{aligned} \int_G E_P\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n) | \mathcal{G}_{t,T}\right] dP &= \int_G \lim_{n \rightarrow \infty} V_T^I(\psi^n) dP \\ &= \int_G \limsup_{n \rightarrow \infty} \mathbf{1}_G V_T^I(\psi^n) dP \\ &\geq \limsup_{n \rightarrow \infty} \int_G V_T^I(\psi^n) dP = \limsup_{n \rightarrow \infty} \int_G E_P[V_T^I(\psi^n) | \mathcal{G}_{t,T}] dP. \end{aligned}$$

Hence, we obtain as above

$$E_{Q \odot P}\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n)\right] \geq \sum_{t < T} E_{Q \odot P}[\gamma_t X_{t,T}] \quad (26)$$

and the claim follows. $\square$

Proof of Theorem 3.6. (i) Assume that (21) holds and that we had an insurance-finance arbitrage, i.e.

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \in L_0^+ \setminus \{0\}$$

for some insurance portfolio strategy $(\psi^n)_{n \geq 1} = (\psi_t^n)_{n \geq 1, t < T}$ and some financial strategy $(\xi_t)_{t \leq T}$. First, for any $Q \in \mathcal{M}_e(\mathbb{F})$, $E_{Q \odot P}[V_T^F(\xi)] = E_Q[V_T^F(\xi)] = 0$ by Proposition 2.1 (i). Then,

$$\begin{aligned} E_{Q \odot P}\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi)\right] &= E_{Q \odot P}\left[\lim_{n \rightarrow \infty} V_T^I(\psi^n)\right] \\ &= E_{Q \odot P}\left[\lim_{n \rightarrow \infty} \sum_{t=0}^{T-1} \sum_{i \geq 1} \psi_t^i (p_t - X_{t,T}^i)\right]. \end{aligned} \quad (27)$$

With the equations (23), (24) we obtain that

$$(27) = \sum_{t=0}^{T-1} E_{Q \odot P}\left[\gamma_t (p_t - E_{Q \odot P}[X_{t,T} | \mathcal{F}_T])\right].$$

We consider the specific Q from (21). Then, $p_t \leq E_{Q \odot P}[X_{t,T} | \mathcal{F}_t]$, P -a.s., hence

$$(27) \leq \sum_{t=0}^{T-1} E_{Q \odot P}\left[\gamma_t (E_{Q \odot P}[X_{t,T} | \mathcal{F}_T] - E_{Q \odot P}[X_{t,T} | \mathcal{F}_t])\right] = 0,$$

a contradiction to the assumption of an insurance-finance arbitrage.

(ii) Assume that (22) holds. We set,

$$A_t := \bigcap_{Q \in \mathcal{M}_e(\mathbb{F})} \{p_t^\downarrow > E_{Q \odot P}[X_{t,T} | \mathcal{F}_t]\} \in \mathcal{F}_t$$

and, by assumption, $P(A_t) > 0$.

This implies that

$$\mathbf{1}_{A_t} p_t \geq \mathbf{1}_{A_t} p_t^\downarrow \geq \mathbf{1}_{A_t} \operatorname{ess\,sup}_{Q \in \mathcal{M}_e(\mathbb{F})} E_{Q \odot P}[X_{t,T} | \mathcal{F}_t] = \mathbf{1}_{A_t} \operatorname{ess\,sup}_{Q \in \mathcal{M}_e(\mathbb{F})} E_Q\left[E_P[X_{t,T} | \mathcal{G}_{t,T}] | \mathcal{F}_t\right] =: \pi_t. \quad (28)$$

At t we take the uniform allocation $\psi_t^n = \mathbb{1}_{A_t}(n^{-1}, \dots, n^{-1}, 0, \dots)$ over the first n insurance seekers, restricted to the set A_t and $\psi_s^n = 0$ for all $s \neq t < T$. This is an admissible strategy and since

$$\sum_{i \geq 1} \frac{1}{i^2} \text{Var}(X_{t,T}^i | \mathcal{G}_{t,T}) = \text{Var}(X_{t,T} | \mathcal{G}_{t,T}) \sum_{i \geq 1} \frac{1}{i^2} < \infty,$$

we are entitled to apply the conditional strong law of large numbers given in Theorem 3.5 in [Majerek et al. \(2005\)](#). Hence with Assumption 3.1 and the fact that $\gamma_t = \sum_{i \geq 1} \psi_t^{n,i} = \mathbb{1}_{A_t} \in \mathcal{F}_t$, we get

$$\sum_{i \geq 1} \psi_t^{n,i} X_{t,T}^i \rightarrow \mathbb{1}_{A_t} E_P[X_{t,T} | \mathcal{G}_{t,T}] =: H, \quad (29)$$

P -almost surely as $n \rightarrow \infty$. Therefore

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) = \mathbb{1}_{A_t} (p_t - E_P[X_{t,T} | \mathcal{G}_{t,T}]) \geq \mathbb{1}_{A_t} (p_t^\downarrow - H).$$

As π_t is the conditional superhedging price of H , we obtain from Theorem 7.2 in [Föllmer and Schied \(2004\)](#)⁷ that there is a superhedging strategy $\xi = \mathbb{1}_{A_t} \xi$ such that

$$\mathbb{1}_{A_t} \left(\pi_t + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s \right) \geq H. \quad (30)$$

Using this financial trading strategy ξ , we find from (28) that

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \geq \mathbb{1}_{A_t} (p_t^\downarrow - H + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s) \quad (31)$$

almost surely. Now, using (28) and (30), we obtain

$$(31) \geq \mathbb{1}_{A_t} (\pi_t - \pi_t) = 0. \quad (32)$$

For the final step we have to distinguish if the claim H is replicable or not. For the first case let

$$B_t := \left\{ \text{ess sup}_{Q \in \mathcal{M}_e(\mathbb{F})} E_Q[H | \mathcal{F}_t] = \text{ess inf}_{Q \in \mathcal{M}_e(\mathbb{F})} E_Q[H | \mathcal{F}_t] \right\}$$

and assume $P(A_t \cap B_t) > 0$. By assumption (22), we have $p_t^\downarrow > \pi_t$ on a set of positive probability. This allows to drop equality in (32): indeed, since $p_t^\downarrow > \pi_t$ with positive probability, we obtain from (31) that

$$(31) \geq \mathbb{1}_{A_t \cap B_t} (p_t^\downarrow - H + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s) > \mathbb{1}_{A_t \cap B_t} (\pi_t - \pi_t) = 0, \quad (33)$$

with positive probability. Hence, $\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \neq 0$ and therefore

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \in L_+^0 \setminus \{0\}.$$

For the second case let

$$B'_t := A_t \cap \left\{ \text{ess sup}_{Q \in \mathcal{M}_e(\mathbb{F})} E_Q[H | \mathcal{F}_t] > \text{ess inf}_{Q \in \mathcal{M}_e(\mathbb{F})} E_Q[H | \mathcal{F}_t] \right\}$$

and assume $P(A_t \cap B'_t) > 0$. Again, we can drop equality in (32): indeed, we obtain analogously that

$$(31) \geq \mathbb{1}_{A_t \cap B'_t} (p_t^\downarrow - H + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s) \geq \mathbb{1}_{A_t \cap B'_t} (\pi_t - H + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s) \geq 0. \quad (34)$$

Since on B'_t , the no-arbitrage interval of the European claim H is a true interval, the upper bound of the conditional no-arbitrage interval, π_t , already yields a (financial) arbitrage (on B'_t). Hence, $\mathbb{1}_{A_t \cap B'_t} (\pi_t - H + \xi \Delta S) \in L_+^0 \setminus \{0\}$, and, in addition,

$$\lim_{n \rightarrow \infty} V_T^I(\psi^n) + V_T^F(\xi) \in L_+^0 \setminus \{0\}.$$

The existence of an insurance-finance arbitrage is proved. $\square$

⁷Compare Theorem 2.4.4. in [Delbaen and Schachermayer \(2006\)](#), whose proof however requires a finite Ω .

Remark 3.8 (Using the QP-rule for arbitrage-free valuation). First, the measure $\mathbb{Q} \odot \mathbb{P}$ constructed in Proposition 2.1 can directly be used for valuation excluding financial arbitrage, since the restriction $\mathbb{Q} \odot \mathbb{P}|_{\mathcal{F}_T}$ is an equivalent $\mathbb{F}$ -martingale measure.

Second, in the context of Theorem 3.6, the QP-rule is the main tool to exclude insurance-finance arbitrage. Since it is also market-consistent as observed in Section 2.3, it comes as a candidate for the valuation of insurance contracts, in particular for the valuation of hybrid contracts depending on the financial market. $\diamond$

Example 3.9 (Valuation by the QP-rule). We continue in the setting of example 2.2: consider the different insurance claims $X = \mathbb{1}_{\{p\}}$ with $p = \{(g, p), (m, p), (b, p)\}$ and $Y = XC$ where C is the call on S at strike 0.7.

Complete market: if $k = 0$ and $h, l \in (0, 1)$, then $Q(g) = 1/2$ and we obtain

$$E_{\mathbb{Q} \odot \mathbb{P}}[\mathbb{1}_{\{(g, p)\}}] = 1/2 \cdot 0.1 = 0.05.$$

Hence, $E_{\mathbb{Q} \odot \mathbb{P}}[X] = 0.05 + 0.2 = 0.25$ and every value $p \leq 0.25$ does not allow for an insurance-finance arbitrage.

Incomplete market: if $k, h, l \in (0, 1)$, then

$$E_{\mathbb{Q} \odot \mathbb{P}}[\mathbb{1}_{\{(g, p)\}}] = s \cdot 0.1.$$

Then, $E_{\mathbb{Q} \odot \mathbb{P}}[X] = 0.1s + 0.2(1 - 2s) + 0.4s = 0.2 + 0.1s \in (0.2, 0.25)$. Hence, every value $p < 0.25$ does not allow for an insurance-finance arbitrage. As for Y we find $E_{\mathbb{Q} \odot \mathbb{P}}[Y] = 0.1s(1.5 - 0.7) + 0.2(1 - 2s)(1 - 0.7) = 0.06 - 0.04s$. Thus, a premium for the hybrid contract with benefits Y must be less than 0.06 if insurance arbitrage were excluded. $\diamond$

Remark 3.10 (On the role of $p^\downarrow$). From a general perspective, the premium p_t should be a $\mathcal{G}_t$ -measurable random variable. However, the knowledge of p_t might then lead to arbitrage possibilities on the financial ($\mathbb{F}$ -adapted) market. This is the reason why we introduced $p^\downarrow$ in the theorem above.

In the practically most relevant cases the setting can be simplified as follows: Enlarge the filtration $\mathbb{F}$ progressively by p leading to the filtration $\tilde{\mathbb{F}} = (\tilde{\mathcal{F}}_t)_{t \in \mathbb{T}}$ with

$$\tilde{\mathcal{F}}_t = \mathcal{F}_t \vee \sigma\{p_s, s \leq t\}. \quad (35)$$

The assumption that the financial market with respect to $\tilde{\mathbb{F}}$ is free of financial arbitrage will be satisfied in most applications: meaning that the knowledge of p_t does not introduce arbitrages. Indeed, it is in most cases easy to find premiums of insurance contracts in the Internet. Then, $\mathbb{F}$ can be replaced by $\tilde{\mathbb{F}}$ and p can be considered to be adapted to $\mathbb{F}$.

3.3. Scheduled premiums. Up to now we were assuming that the premium is paid at initiation of the contract. In practice, one typically pays premiums at scheduled time until a certain stopping time. We show in this subsection how to incorporate the scheduled premiums in our setting.

In this regard, let A be a deterministic function of finite variation over $[t, T]$ decoding the payments. For example, the premiums $p^1, \dots, p^n$ at times $t^1, \dots, t^n$ are captured via $A(t) = \sum_{i=1}^n \mathbb{1}_{\{t_i \leq t\}} p^i$.

Consider a $\mathbb{G}$ -stopping time τ at which the payment of the premiums is terminated. The total premiums sum to $A(T)$, while the realized premiums are only $A(\tau \wedge T-)$.

Consider an insurance contract with benefits $X_{t,T}$. Our goal is compute a suitable A or to check, for a given A , if NIFA holds. This can be traced back to the setting of a single premium by the following two steps: first, the initial premium is set to

$$p_t^A = A(T).$$

Second, the insured has the additional option to cease payments after τ , which we simply add to the benefits. In this regard, define the modified benefits

$$X_{t,T}^{A,\tau} = X_{t,T} + (A(T) - A(\tau \wedge T)).$$

Now, Theorem 3.6 can be applied to the premium p_t^A and the benefits $X_{t,T}^{A,\tau}$. Then, the QP-rule in the context of Remarks 3.8 and 3.10 can be applied to compute the basic parts of the scheduled premiums A .

3.4. Adding a risk margin. As already mentioned, premiums in practice consist of a basic part and a part covering risk margins (and possibly other costs), see also the discussion in [Chen et al. \(2020\)](#). We elaborate shortly on the non-linear valuation of the risk margin here. Consider the time $0 \leq t < T$ and consider the allocation ψ_t^n insurance seekers leading to the commitment

$$\sum_{i \geq 1} \psi_t^{i,n} X_{t,T}^i.$$

We decompose the commitment

$$\begin{aligned} \sum_{i \geq 1} \psi_t^{i,n} X_{t,T}^i &= \gamma_t E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_t] \\ &+ \gamma_t \left(E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_{t,T}] - E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_t] \right) \\ &+ \sum_{i \geq 1} \psi_t^{i,n} X_{t,T}^i - \gamma_t E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_{t,T}] =: \gamma_t E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_t] + Y_t^1 + Y_t^{2,n}. \end{aligned}$$

The risk in Y_t^1 can further be reduced by hedging on the financial market: assume that ξ is the chosen hedging strategy, then the remaining risk is given by

$$\tilde{Y}_t^1 = Y_t^1 + \sum_{s=t}^{T-1} \xi_s \cdot \Delta S_s.$$

When we have a complete market or a perfect hedge, $\tilde{Y}_t^1 = 0$.

The risk margin can be computed by using a time-consistent risk measure which we denote by $(\rho_t)_{t \in \mathbb{T}}$. In this case, the *security or supervisory margin* is given by

$$\rho_t(\tilde{Y}_t^1 + Y_t^{2,n}) \quad (36)$$

and the *cost-of-capital part* of it, i.e. the *risk margin*, has to be added to the basic part of the premium $E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_t]$.

In the case where a conditional strong law of large numbers applies, which is for example used in Theorem 3.6, then $Y_t^{2,n} \rightarrow 0$ almost surely: a part of the insurance risk can be diversified away by choosing an appropriately large portfolio and the security margin simplifies to $\rho_t(\tilde{Y}_t^1)$.

An alternative decomposition of the premium into the terms $\gamma_t E_{\mathbb{Q} \odot \mathbb{P}}[X_{t,T} | \mathcal{G}_t] + \rho_t(Y_t^1) + \rho_t(Y_t^{2,n})$ has recently been proposed in [Deelstra et al. \(2018\)](#), however under stronger assumptions on the financial market and the choice of the equivalent martingale measure.

4. PROGRESSIVE ENLARGEMENT OF THE FINANCIAL FILTRATION

Besides the publicly available information $\mathbb{F}$, the insurer has access to private information, given by the filtration $\mathbb{G}$. In this section we introduce additional structure on $\mathbb{G}$ by utilizing the theory of progressive enlargements.

Consider a random time τ which could be used, e.g., to model the random life time of the insured, a claim arrival time, or a surrender time. We consider multiple times in Section 4.1.

We require that $\{\tau > t\}$, $t \in \mathbb{T}$, are atoms in the filtration $\mathbb{G}$; more precisely, we assume that for all $t \in \mathbb{T}$,

$$\mathcal{G}_t \cap \{\tau > t\} = \mathcal{F}_t \cap \{\tau > t\}, \quad (37)$$

which means that for each $G \in \mathcal{G}_t$ there exists an $F \in \mathcal{F}_t$, such that $G \cap \{\tau > t\} = F \cap \{\tau > t\}$ and vice versa.

A classical and well-known example where (37) holds is the progressive enlargement of $\mathbb{F}$ with the random time τ . The enlarged filtration is the smallest filtration such that τ becomes a stopping time.

Remark 4.1 (Progressive enlargement). Assume that for all $t \in \mathbb{T}$,

$$\mathcal{G}_t := \sigma(\{F \cap \{\tau \leq s\} : F \in \mathcal{F}_t, s \leq t\}) =: \mathcal{F}_t \vee \{\tau \wedge t\}, \quad (38)$$

then $\mathbb{G} = (\mathcal{G}_t)_{t \geq 0}$ is called the *progressive enlargement* of $\mathbb{F}$ with τ . Then (37) holds: indeed, this follows from

$$F \cap \{\tau > s\} \cap \{\tau > t\} = F \cap \{\tau > t\},$$

for $0 \leq s \leq t \leq T$ and $F \in \mathcal{F}_t$. For a detailed study and many references to related literature see [Aksamit and Jeanblanc \(2017\)](#). Typically we will be interested in a filtration which contains more information, like the employment status, health status, surrender behaviour, etc. $\diamond$

We place ourselves in the context of Remark 3.10. To this end, fix t and assume that the information $\mathcal{G}_t$ does not generate arbitrage, i.e. there exists an equivalent martingale measure Q on $(\Omega, \mathcal{G}_{t,T})$. The QP-rule now works as follows: the insurance premium p_t is computed via taking expectations under $Q \odot P$ with respect to $\tilde{\mathcal{F}}_t = \mathcal{G}_t$. Denote by

$$G_t = Q \odot P(\tau > t | \mathcal{F}_t), \quad t \geq 0 \quad (39)$$

the Azéma supermartingale. The Azéma supermartingale decodes the survival probability of τ in the small filtration $\mathbb{F}$.

Proposition 4.2. Assume $\mathbb{F} \subset \mathbb{G}$, τ being a $\mathbb{G}$ -stopping time, and that (37) holds. Then, for any $\mathbb{G}$ -adapted process A bounded from below,

$$E_{Q \odot P}[A_\tau | \mathcal{G}_t] = \mathbf{1}_{\{\tau \leq t\}} A_\tau + \mathbf{1}_{\{\tau > t, G_t > 0\}} G_t^{-1} E_Q \left[E_P[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{F}] | \mathcal{F}_t \right] \quad (40)$$

holds for all $t \in \mathbb{T}$.

Proof. First, we consider $A_\tau = A_\tau \mathbf{1}_{\{\tau \leq t\}} + A_\tau \mathbf{1}_{\{\tau > t\}}$. Since $A_\tau \mathbf{1}_{\{\tau \leq t\}}$ is $\mathcal{G}_t$ -measurable, we obtain $E_{Q \odot P}[A_\tau \mathbf{1}_{\{\tau \leq t\}} | \mathcal{G}_t] = A_\tau \mathbf{1}_{\{\tau \leq t\}}$, the first addend of (40).

Second, we consider $A_\tau \mathbf{1}_{\{\tau > t\}}$. Note that $E_{Q \odot P}[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t]$ vanishes on the set $\{\tau > t, G_t = 0\}$, and so does the second addend in (40). So, we may assume $G_t > 0$ everywhere. Now, (37) implies that for any $t \geq 0$ and any $\mathcal{G}_t$ -measurable random variable $\tilde{A}_t$ we find a $\mathcal{F}_t$ -measurable random variable Y_t , such that

$$\tilde{A}_t \mathbf{1}_{\{\tau > t\}} = Y_t \mathbf{1}_{\{\tau > t\}} \quad (41)$$

by an application of the monotone class theorem. Using $E_{Q \odot P}[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{G}_t]$ for $\tilde{A}_t$ in this equation and taking conditional expectation with respect to $\mathcal{F}_t$, we obtain that

$$G_t^{-1} E_{Q \odot P} \left[E_{Q \odot P}[A_\tau | \mathcal{G}_t] \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t \right] = G_t^{-1} E_{Q \odot P} \left[Y_t \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t \right] = Y_t. \quad (42)$$

On the other hand,

$$\begin{aligned} E_{Q \odot P} \left[E_{Q \odot P}[A_\tau | \mathcal{G}_t] \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t \right] &= E_{Q \odot P} [A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{F}_t] \\ &= E_{Q \odot P} \left[E_P[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{F}] | \mathcal{F}_t \right] \\ &= E_Q \left[E_P[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{F}] | \mathcal{F}_t \right], \end{aligned} \quad (43)$$

where we have used that, according to Proposition 2.1, $Q \odot P$ coincides with P , conditionally on $\mathcal{F}$, and, that it coincides with Q on $(\Omega, \mathcal{F})$. This shows (40). $\square$

Example 4.3. We continue Example 2.3 and compute the above quantities in this setting. Recall that the cumulated intensity Λ was an $\mathbb{F}$ -adapted, increasing process and $P(\tau > t | \mathcal{F}_t) = e^{-\Lambda_t}$ and that A was $\mathbb{F}$ -adapted. We obtained that

$$P(\tau > t | \mathcal{F}) = P(\Lambda_t < E | \mathcal{F}) = e^{-\Lambda_t}.$$

On the other side,

$$e^{-\Lambda_t} = Q \odot P(\tau > t | \mathcal{F}_t) = G_t.$$

Moreover,

$$\begin{aligned} E_{Q \odot P}[A_\tau \mathbf{1}_{\{\tau > t\}} | \mathcal{G}_t] &= \mathbf{1}_{\{\tau \leq t\}} A_\tau + \mathbf{1}_{\{\tau > t\}} G_t^{-1} \sum_{s=t+1}^T E_Q[A_s P(\tau = s | \mathcal{F}) | \mathcal{F}_t] \\ &= \mathbf{1}_{\{\tau \leq t\}} A_\tau + \mathbf{1}_{\{\tau > t\}} e^{\Lambda_t} \sum_{s=t+1}^T E_Q[A_s (e^{-\Lambda_{s-1}} - e^{-\Lambda_s}) | \mathcal{F}_t]. \end{aligned} \quad (44)$$

This formula is a key result for the valuation of a large number of hybrid products. $\diamond$

An additional difficulty arises when A is not $\mathbb{F}$ -measurable. Here one is able to exploit the structure of the progressive enlargement in the case where τ is honest. For example, Corollary 5.12 in [Aksamit and Jeanblanc \(2017\)](#) allows to decompose X in several $\mathbb{F}$ -adapted components on random intervals depending solely on τ .

Proposition 4.4. *Assume $\mathbb{F} \subset \mathbb{G}$, τ being a $\mathbb{G}$ -stopping time, and that (37) holds. Then, for any $\mathbb{G}$ -adapted process A and any $\mathcal{F}_T$ -measurable random variable F_T , both bounded from below, it holds that*

$$\begin{aligned} E_{\mathbb{Q} \odot \mathbb{P}}[F_T A_\tau | \mathcal{G}_t] &= \mathbb{1}_{\{\tau \leq t\}} A_\tau E_{\mathbb{Q} \odot \mathbb{P}}[F_T | \mathcal{G}_t] \\ &\quad + \mathbb{1}_{\{\tau > t, G_t > 0\}} G_t^{-1} E_Q \left[F_T E_P[A_\tau \mathbb{1}_{\{\tau > t\}} | \mathcal{F}] | \mathcal{F}_t \right] \end{aligned}$$

holds for all $t \in \mathbb{T}$.

The proof follows as for Proposition 4.2.

4.1. Multiple stopping times. For the insurance company it is of course important to consider more than one stopping time, and to allow for dependence between these stopping times. We generalize the previous framework by considering a countable number of atoms and provide the associated generalizations of the previously obtained results.

Consider for each $t \in \mathbb{T}$ $\mathcal{G}_t$ -measurable sets $(P_t^1, P_t^2, \dots, P_t^n)$ such that $P_t^i \cap P_t^j = \emptyset$ for $i \neq j$. Assume that for all $1 \leq i \leq n$ and $t \in \mathbb{T}$,

$$\mathcal{G}_t \cap P_t^i = \mathcal{F}_t \cap P_t^i \quad (45)$$

and denote by

$$G_t^i = \mathbb{Q} \odot \mathbb{P}(P_t^i | \mathcal{F}_t) \quad (46)$$

the respective generalization of the Azéma supermartingale. Note that without additional assumptions G^i does not need to be a supermartingale. Nevertheless, we have the following generalization of Proposition 4.2. Set $\Omega_t = \sum_{i=1}^n P_t^i$.

Proposition 4.5. *Assume $\mathbb{F} \subseteq \mathbb{G}$, and that (45) and (46) hold. Then, for any $\mathcal{G}$ -measurable random variable A bounded from below,*

$$\mathbb{1}_{\Omega_t} E_{\mathbb{Q} \odot \mathbb{P}}[A | \mathcal{G}_t] = \sum_{i=1}^n \mathbb{1}_{P_t^i \cap \{G_t^i > 0\}} (G_t^i)^{-1} E_Q \left[E_P[A \mathbb{1}_{P_t^i} | \mathcal{F}] | \mathcal{F}_t \right] \quad (47)$$

holds for all $t \in \mathbb{T}$.

Proof. First, we decompose

$$\mathbb{1}_{\Omega_t} E_{\mathbb{Q} \odot \mathbb{P}}[A | \mathcal{G}_t] = \sum_{i=1}^n \mathbb{1}_{P_t^i} E_{\mathbb{Q} \odot \mathbb{P}}[A \mathbb{1}_{P_t^i} | \mathcal{G}_t]. \quad (48)$$

For the following, we fix i . From (45), a monotone class arguments gives the following: for a $\mathcal{G}_t$ -measurable random variable X_t we can find a $\mathcal{F}_t$ -measurable random variable $\tilde{Y}_t$, such that

$$X_t \mathbb{1}_{P_t^i} = \tilde{Y}_t \mathbb{1}_{P_t^i}. \quad (49)$$

Hence, there exists an $\mathcal{F}_t$ -measurable random variable Y_t such that

$$E_{\mathbb{Q} \odot \mathbb{P}}[A | \mathcal{G}_t] \mathbb{1}_{P_t^i} = Y_t \mathbb{1}_{P_t^i}. \quad (50)$$

Taking conditional expectations with respect to $\mathcal{F}_t$ and multiplying with $\mathbb{1}_{P_t^i}$ yields that

$$\mathbb{1}_{P_t^i} E_{\mathbb{Q} \odot \mathbb{P}} \left[E_{\mathbb{Q} \odot \mathbb{P}}[A \mathbb{1}_{P_t^i} | \mathcal{G}_t] | \mathcal{F}_t \right] = Y_t G_t^i \mathbb{1}_{P_t^i}.$$

The left hand side equals $\mathbb{1}_{P_t^i} E_{\mathbb{Q} \odot \mathbb{P}}[A \mathbb{1}_{P_t^i} | \mathcal{F}_t]$, which consequently vanishes when $G_t^i = 0$. Inserting (50), we obtain that

$$\mathbb{1}_{P_t^i \cap \{G_t^i > 0\}} (G_t^i)^{-1} E_{\mathbb{Q} \odot \mathbb{P}}[A \mathbb{1}_{P_t^i} | \mathcal{F}_t] = E_{\mathbb{Q} \odot \mathbb{P}}[A \mathbb{1}_{P_t^i} | \mathcal{G}_t] \mathbb{1}_{P_t^i}.$$

The claim now follows by combining this with Proposition 2.1 and (48). $\square$

4.2. Two stopping times. The for us most interesting case is the case of two stopping times, which will be the random life time of the insured and the surrender time. Typically it is assumed that they are conditionally independent. This can be a serious restriction for the applications we have in mind: indeed, dependence between remaining life time and surrender is of course possible and should be taken into account. We may use the above result to do so. Motivated by this, we will develop Proposition 4.5 further in the case of two stopping times.

Consider two $\mathbb{F}$ -adapted, increasing processes Λ^1 , and Λ^2 . Assume there exists two standard exponential random variables E^1 , and E^2 with continuous copula

$$C(u_1, u_2) := P(\exp(-E^1) < u_1, \exp(-E^2) < u_2).$$

Let $\sigma = \inf\{t \in \mathbb{T} : \Lambda_t^1 \geq E^1\} \wedge T$, and $\tau = \inf\{t \in \mathbb{T} : \Lambda_t^2 \geq E^2\} \wedge T$. Using independence of E^i and $\mathcal{F} = \mathcal{F}_T$, we obtain that

$$\begin{aligned} P(\sigma > s, \tau > t | \mathcal{F}) &= P(\Lambda_s^1 < E^1, \Lambda_t^2 < E^2 | \mathcal{F}) \\ &= P(\exp(-E^1) < \Lambda_s^1, \exp(-E^2) < \Lambda_t^2 | \mathcal{F}) = C(e^{-\Lambda_s^1}, e^{-\Lambda_t^2}). \end{aligned} \quad (51)$$

Example 4.6 (Conditional independence). If in addition, E_1 and E_2 are independent, then σ and τ are independent conditional on $\mathcal{F}$ and $C(u, v) = u \cdot v$. This simplifies the computations significantly since then $P(\sigma > s, \tau > t | \mathcal{F}) = \exp(-\Lambda_s^1 - \Lambda_t^2)$. $\diamond$

Fix $t \in \mathbb{T}$ and consider the disjoint $\mathcal{G}_t$ -measurable sets

$$\begin{aligned} P_t^{1,1} &= \{\sigma > t, \tau > t\}, \\ P_t^{2,i} &= \{\sigma > t, \tau = i\}, \quad i = 0, \dots, t-1, \\ P_t^{3,i} &= \{\sigma = i, \tau > t\}, \quad i = 0, \dots, t-1, \\ P_t^{4,i,j} &= \{\sigma = i, \tau = j\}, \quad i, j = 0, \dots, t-1, \end{aligned}$$

which will take the role of $P_t^1, \dots, P_t^n$ in the previous section. Let $\Gamma(s, t) = C(\exp(-\Lambda_s^1), \exp(-\Lambda_t^2))$. Then we obtain from (51) that

$$\begin{aligned} G_t^{1,1} &= Q \odot P(\sigma > t, \tau > t | \mathcal{F}_t) = \Gamma(t, t), \\ G_t^{2,i} &= Q \odot P(\sigma > t, \tau = i | \mathcal{F}_t) = \Gamma(t, i-1) - \Gamma(t, i), \\ G_t^{3,i} &= Q \odot P(\sigma = i, \tau > t | \mathcal{F}_t) = \Gamma(i-1, t) - \Gamma(i, t), \\ G_t^{4,i,j} &= Q \odot P(\sigma = i, \tau = j | \mathcal{F}_t) \\ &= \Gamma(i-1, j-1) - \Gamma(i, j-1) - \Gamma(i-1, j) + \Gamma(i, j). \end{aligned}$$

Now, consider two $\mathbb{F}$ -adapted payment streams A^1 and A^2 . If the insurer dies at σ before surrendering, he will receive A_σ^1 while if he first surrenders, he will receive A_τ^2 . Precisely, this defines the following payoff:

$$X_{t,T} = \mathbb{1}_{\{\sigma > t, \tau > t\}} (\mathbb{1}_{\{\sigma < T, \sigma \leq \tau\}} A_\sigma^1 + \mathbb{1}_{\{\tau < T, \tau < \sigma\}} A_\tau^2). \quad (52)$$

Proposition 4.7. For the payoff in (52) we have that

$$E_{Q \odot P}[X_{t,T} | \mathcal{G}_t] = \mathbb{1}_{\{\sigma > t, \tau > t\}} (G_t^{1,1})^{-1} \sum_{s=t+1}^{T-1} E_Q[A_s^1 G_{s-1}^{3,s} + A_s^2 G_s^{2,s} | \mathcal{F}_t].$$

Proof. For the part with A^1 we obtain the following decomposition:

$$\begin{aligned} \mathbb{1}_{\{t < \sigma < T, \sigma \leq \tau\}} A_\sigma^1 &= \sum_{s=t+1}^{T-1} \mathbb{1}_{\{\sigma = s, \tau \geq s\}} A_s^1 = \sum_{s=t+1}^{T-1} \mathbb{1}_{\{\sigma = s, \tau > s-1\}} A_s^1 \\ &= \sum_{s=t+1}^{T-1} P_{s-1}^{3,s} A_s^1. \end{aligned}$$

Similarly,

$$\mathbb{1}_{\{\tau < T, \tau < \sigma\}} A_\tau^2 = \sum_{s=t+1}^{T-1} P_s^{2,s} A_s^2.$$

The result now follows by applying Proposition 4.5. $\square$

5. AN AFFINE FRAMEWORK

Affine processes are a well-established tool for modeling term structures, in particular stochastic mortality term structures. We refer to Keller-Ressel et al. (2019) for a detailed treatment of affine processes, including affine processes in discrete time, and to Biffis (2005), Schrager (2006), Luciano et al. (2008), and Blackburn and Sherris (2013) for applications to stochastic mortality. In this section we propose a framework in discrete time for the valuation of insurance products linked to financial markets.

We now assume that there is a driving d' -dimensional process Z , which is affine. The process Z is called *affine*, if it is a Markov process and its characteristic function is exponential affine. We additionally require the existence of exponential moments, such that

$$E[\exp(uZ_{t+1})|Z_t] = \exp(A(u) + B(u) \cdot Z_t)$$

for $0 \leq t < T$ and all $u \in \mathbb{R}^{d'}$; with deterministic functions $A : \mathbb{R}^{d'} \rightarrow \mathbb{R}$ and $B : \mathbb{R}^{d'} \rightarrow \mathbb{R}^{d'}$. We will denote the state space of Z by $\mathcal{Z}$. Typically $\mathcal{Z} = \mathbb{R}_{\geq 0}^m \times \mathbb{R}^n$ with $m + n = d'$.

In the light of the QP-rule it will be important to distinguish between the $\mathbb{F}$ -adapted parts of Z and the parts which are only $\mathbb{G}$ -adapted. This will become important when we consider the set of equivalent martingale measures $\mathcal{M}_e(\mathbb{F})$ where the choice of $\mathbb{F}$ plays an important role. In this regard, let $Z = (X, Y)$ with Y generating the filtration $\mathbb{F}$; here X is d_1 -dimensional and Y is d_2 -dimensional, with $d_1 + d_2 = d'$. We will allow that either d_1 or d_2 vanishes. In the light of Remark 3.10, X might contain insurance-related quantities which are not publicly available.

We model the stock market as exponential driven by the affine process with an additional drift, i.e. we assume that discounted stock prices are given by

$$S_t = \exp(a_0 t + a \cdot Y_t), \quad t \geq 0,$$

with $(a_0, a) = (a_0, a_1, \dots, a_{d_2}) \in \mathbb{R}^{1+d_2}$. This modeling contains the Black-Scholes model and exponential Lévy models as a special case (in discrete time).

5.1. The case of one stopping time. For the modeling of τ , we follow the doubly stochastic approach introduced in Example 2.3. Precisely, we consider a non-decreasing process $\Lambda \geq 0$ given by

$$\Lambda_t = b_0 + \sum_{s=0}^t (b \cdot X_s + c \cdot Y_s), \quad t \geq 0$$

with $(b_0, b, c) \in \mathbb{R}^{1+d'}$. Here, b_0 is chosen such that $\Lambda_0 = 0$, guaranteeing in particular $P(\tau = 0) = 0$, and (b, c) are chosen such that Λ is increasing.

Denote by $\mathbb{F}^Z$ the filtration generated by Z . Together with Λ we assume that τ satisfies (7) and that $\mathbb{G}$ is the progressive enlargement of $\mathbb{F}^Z$ with τ , $\mathcal{G}_t = \mathcal{F}_t^Z \vee (\tau \wedge t)$, for all $t \in \mathbb{T}$.

To ensure absence of financial arbitrage, we assume that there exists an equivalent martingale measure Q . For tractability, we assume that Y is again affine under Q (with existing exponential moments),

$$E_Q[\exp(uY_{t+1})|Y_t] = \exp(A_Q(u) + B_Q(u) \cdot Y_t) \quad (53)$$

for all $0 \leq t < T$ and $u \in \mathbb{R}^{d_2}$.

Remark 5.1. The Esscher change of measure is one well-known example which keeps the affine property during the measure change, but not the only one, see for example Kallsen and Muhle-Karbe (2010).

Finally, we assume that the conditional distribution of X_t , conditional on Y_t and X_{t-1} has affine form and denote

$$E_P[\exp(u \cdot X_t)|Y_t, X_{t-1}] = \exp(\alpha(u) + \beta(u) \cdot X_{t-1} + \gamma(u) \cdot Y_t), \quad (54)$$

for all $0 < t \leq T$ and $u \in \mathbb{R}^{d'}$. The coefficients α, β , and γ can be computed from A and B .

We introduce the following recursive notation: define $\phi(T) = \alpha(-b) + A_Q(a - c + \gamma(-b))$, $\psi^1(T) = \beta(-b)$, and $\psi^2(T) = B_Q(a - c + \gamma(-b))$; now let $u(s) = \psi^1(s + 1) - b$, $v(s) = \psi^2(s + 1) - c$ and set

$$\begin{aligned}\phi(s) &= \alpha(u(s)) + A_Q(v(s) + \gamma(u(s))), \\ \psi^1(s) &= \beta(u(s)), \\ \psi^2(s) &= B_Q(v(s) + \gamma(u(s)))\end{aligned}\tag{55}$$

for $s = 0, \dots, T-1$. Moreover, $\phi'(T) = A_Q(a)$, $\psi'^1(T) = 0$, and $\psi'^2(T) = B_Q(a)$; following the same recursion rule (55). We are interested in the claim $S_T \mathbb{1}_{\{t < \tau \leq T\}}$, which we decompose to

$$\sum_{s=t+1}^T S_s (\mathbb{1}_{\{\tau > s-1\}} - \mathbb{1}_{\{\tau > s\}}).$$

The following proposition now allows to value this claim as well as the claim $S_T \mathbb{1}_{\{\tau > T\}}$. Denote

$$\Phi(t, T) = \sum_{s=t+1}^T \phi(s),\tag{56}$$

and, analogously, Φ' . Moreover we write $\psi^1 \cdot X + \psi^2 \cdot Y = \psi \cdot Z$, again analogously for ψ' .

Proposition 5.2. *We have the following valuation-results*

$$\begin{aligned}E_{\mathbb{Q} \otimes \mathbb{P}}[S_T \mathbb{1}_{\{\tau > T\}} \mid \mathcal{G}_t] &= \mathbb{1}_{\{\tau > t\}} e^{a_0 T + \Phi(t, T) + \psi(t+1) \cdot Z_t}, \\ E_{\mathbb{Q} \otimes \mathbb{P}}[S_T \mathbb{1}_{\{\tau > T-1\}} \mid \mathcal{G}_t] &= \mathbb{1}_{\{\tau > t\}} e^{a_0 T + \Phi'(t, T) + \psi'(t+1) \cdot Z_t}.\end{aligned}$$

Proof. Using Proposition 4.4, and the affine representations of S and Λ , we obtain that

$$\begin{aligned}E_{\mathbb{Q} \otimes \mathbb{P}}[S_T \mathbb{1}_{\{\tau > T\}} \mid \mathcal{G}_t] &= \mathbb{1}_{\{\tau > t\}} e^{\Lambda_t} E_{\mathbb{Q} \otimes \mathbb{P}}[S_T e^{-\Lambda_T} \mid \mathcal{F}_t^Z] \\ &= \mathbb{1}_{\{\tau > t\}} E_{\mathbb{Q} \otimes \mathbb{P}} \left[\exp \left(a_0 T + a \cdot Y_T - \sum_{s=t+1}^T (b \cdot X_s + c \cdot Y_s) \right) \mid Z_t \right].\end{aligned}$$

Now we proceed iteratively: first, consider $s = T$. Then,

$$\begin{aligned}E_P[e^{-b \cdot X_T} \mid \mathcal{F}_{T-1}^Z \vee Y_T] &= e^{\alpha(-b) + \beta(-b) \cdot X_{T-1} + \gamma(-b) \cdot Y_T} \\ E_Q[e^{(a-c+\gamma(-b)) \cdot Y_T} \mid \mathcal{F}_{T-1}^Z] &= e^{A_Q(a-c+\gamma(-b)) + B_Q(a-c+\gamma(-b)) \cdot Y_{T-1}}.\end{aligned}$$

In the next step, for $s = T-1$,

$$E_{\mathbb{Q} \otimes \mathbb{P}}[e^{u \cdot X_s + v \cdot Y_s} \mid \mathcal{F}_{s-1}^Z] = e^{\alpha(u) + \beta(u) \cdot X_{s-1} + A_Q(v + \gamma(u)) + B_Q(v + \gamma(u)) \cdot Y_{s-1}}\tag{57}$$

with $u = \beta(-b) - b$ and $v = B_Q(a - c + \gamma(-b)) - c$. Proceeding iteratively until $s = t+1$ yields

$$E_{\mathbb{Q} \otimes \mathbb{P}}[S_T \mathbb{1}_{\{\tau > T\}} \mid \mathcal{G}_t] = \mathbb{1}_{\{\tau > t\}} e^{a_0 T + \sum_{s=t+1}^T \phi(s) + \psi^1(t+1) \cdot X_t + \psi^2(t+1) \cdot Y_t},$$

the first claim. The second claim follows in a similar way. $\square$

5.2. Survival and surrender. Since we are interested in modeling surrender and survival, we need to consider two stopping times. To exploit the full power of the affine framework, we will assume that they are conditionally independent. In this regard, let σ, τ be doubly stochastic random times as introduced in Section 4.2 with associated non-decreasing processes Λ^1 and Λ^2 and under the assumption of Example 4.6. We assume that

$$\Lambda_t^i = b_0^i + \sum_{s=0}^t (b^i \cdot X_s + c^i \cdot Y_s), \quad t \geq 0$$

with $(b_0^i, b^i, c^i) \in \mathbb{R}^{1+d'}$, $i = 1, 2$. Again, the coefficients are chosen such that the processes start in 0 and are increasing.

First, we obtain that under these assumptions

$$\begin{aligned}G_t^{1,1} &= e^{-\Lambda_t^1 - \Lambda_t^2}, \\ G_t^{2,i} &= e^{-\Lambda_t^1} (e^{-\Lambda_{t-1}^2} - e^{-\Lambda_t^2}), \\ G_t^{3,i} &= (e^{-\Lambda_{t-1}^1} - e^{-\Lambda_t^1}) e^{-\Lambda_t^2}\end{aligned}$$

and an analogous expression for G^4 (which we will not use here).

From these expressions it is clear that we need to generalize our previous notions of ϕ and ψ . Fix $0 \leq s < T$ and consider $\kappa_i = (\kappa_i^1, \kappa_i^2) \in \mathbb{R}^{(T+1) \times d'} \times \mathbb{R}^{(T+1) \times d'}$, $0 \leq i < T$ (we use κ^1 for the coefficients associated with X and κ^2 for those associated with Y) defined by $\kappa_T = (-b^1, a - c^1)$, $\kappa_t = (-b^1, -c^1)$ for $s+1 \leq t < T$ and $\kappa_t = (-b^1 - b^2, -c^1 - c^2)$ for $t \leq s$; moreover $\kappa'_T = (-b^2, a - c^2)$, $\kappa'_t = (-b^2, -c^2)$ for $s+1 \leq t < T$ and $\kappa'_t = (-b^1 - b^2, -c^1 - c^2)$ for $t \leq s$.

Define recursively

$$\begin{aligned}\phi(T) &= \alpha(\kappa_T^1) + A_Q(\kappa_T^2 + \gamma(\kappa_T^1)), \\ \psi^1(T) &= \beta(\kappa_T^1), \\ \psi^2(T) &= B_Q(\kappa_T^2 + \gamma(\kappa_T^1)) \\ \phi(i) &= \alpha(u(i)) + A_Q(v(i) + \gamma(u(i))), \\ \psi^1(i) &= \beta(u(i)), \\ \psi^2(i) &= B_Q(v(i) + \gamma(u(i))),\end{aligned}$$

with $u(i) = \psi^1(i+1) + \kappa_i^1$, $v(i) = \psi^2(i+1) + \kappa_i^2$. Again, we write

$$\Phi(t, T) = \sum_{i=t+1}^T \phi(i)$$

and $\psi^1 \cdot X + \psi^2 \cdot Y = \psi \cdot Z$. The coefficients ϕ' , ψ' and Φ' are obtained with the same recursion, replacing κ by κ' .

Proposition 5.3. *We have the following valuation-results for $t \leq s < T$,*

$$\begin{aligned}e^{\Lambda_t^1 + \Lambda_t^2} E_{Q \odot P}[S_T e^{-\Lambda_T^1 - \Lambda_T^2} \mid \mathcal{G}_t] &= e^{a_0 T + \Phi(t, T) + \psi(t+1) \cdot Z_t} \\ e^{\Lambda_t^1 + \Lambda_t^2} E_{Q \odot P}[S_T e^{-\Lambda_T^1 - \Lambda_T^2} \mid \mathcal{G}_t] &= e^{a_0 T + \Phi'(t, T) + \psi'(t+1) \cdot Z_t}\end{aligned}$$

Proof. We proceed iteratively as in the proof of Proposition 5.2. Note that

$$e^{\Lambda_t^1 + \Lambda_t^2} S_T e^{-\Lambda_T^1 - \Lambda_T^2} = e^{a_0 T + a \cdot Y_T - \sum_{i=t+1}^T (b^1 X_i + c^1 Y_i) - \sum_{j=t+1}^s (b^2 X_j + c^2 Y_j)}.$$

For $i = T$ and $i > s$ we obtain with (57)

$$\begin{aligned}E_Q[e^{-b^1 \cdot X_T + (a - c^1) \cdot Y_T} \mid \mathcal{F}_{T-1}] &= e^{\alpha(-b^1) + \beta(-b^1) \cdot X_{T-1} + A_Q(a - c^1 + \gamma(-b^1)) + B_Q(a - c^1 + \gamma(-b^1)) \cdot Y_{T-1}} \\ &= e^{\phi(T) + \psi^1(T) \cdot X_{T-1} + \psi^2(T) \cdot Y_{T-1}}.\end{aligned}$$

For $i \leq T-1$ and $i > s$,

$$E_Q[e^{(\psi^1(i+1) - b^1) \cdot X_i + (\psi^2(i+1) - c^1) \cdot Y_i} \mid \mathcal{F}_{i-1}] = e^{\phi(i) + \psi^1(i) \cdot X_{i-1} + \psi^2(i) \cdot Y_{i-1}}.$$

For $i \leq s$ we have to compute

$$E_Q[e^{(\psi^1(i+1) - b^1 - b^2) \cdot X_i + (\psi^2(i+1) - c^1 - c^2) \cdot Y_i} \mid \mathcal{F}_{i-1}] = e^{\phi(i) + \psi^1(i) \cdot X_{i-1} + \psi^2(i) \cdot Y_{i-1}}$$

and the first claim follows. The second claim is obtained similarly. $\square$

6. EXAMPLES

In this section we provide several examples which illustrate the application of our results.

6.1. The surrender option. A simple example where the full generality of our approach should be used is a surrender option. We ignore for simplicity survival. If the surrender options is exercised, the insured will receive the amount of the underlying portfolio. The surrender time is a random time, and the likelihood for surrender will depend on the performance of the underlying portfolio, see for example Ballotta et al. (2019) and references therein.

Let τ denote the surrender time and consider the affine framework in Section 5.1. Let $t < T$ denote the current time. The payoff of the surrender option is

$$\mathbb{1}_{\{t < \tau < T\}} S_\tau,$$

payments in the case of no surrender can be priced in a similar way and are therefore ignored here. With the aid of Proposition 5.2 its value computes to

$$\begin{aligned} E_{Q \odot P}[\mathbb{1}_{\{t < \tau < T\}} S_\tau | \mathcal{G}_t] &= \mathbb{1}_{\{\tau > t\}} \sum_{i=t+1}^{T-1} E_{Q \odot P}[S_i(\mathbb{1}_{\{\tau > i-1\}} - \mathbb{1}_{\{\tau > i\}}) | \mathcal{G}_t] \\ &= \mathbb{1}_{\{\tau > t\}} \sum_{i=t+1}^{T-1} e^{a_0 i} \left(e^{\Phi'(t,i) + \psi'(t+1) \cdot Z_t} - e^{\Phi(t,i) + \psi(t+1) \cdot Z_t} \right). \end{aligned} \quad (58)$$

6.2. Variable annuities. For the case of a variable annuity we have to consider surrender and survival: let σ be the surrender time and τ the life time of the insured. As a typical payoff illustrating the application of our methodology to the valuation of variable annuities we consider the surrender option in this context: at surrender at σ , the insured receives S_σ , provided $\sigma \leq \tau$. We obtain that

$$\begin{aligned} E_{Q \odot P}[\mathbb{1}_{\{t < \sigma < T, \sigma \leq \tau\}} S_\sigma | \mathcal{G}_t] &= \mathbb{1}_{\{\sigma > t, \tau > t\}} e^{\Lambda_t^1 + \Lambda_t^2} \sum_{s=t+1}^{T-1} E_Q[S_s G_{s-1}^{3,s} | \mathcal{F}_t] \\ &= \mathbb{1}_{\{\sigma > t, \tau > t\}} e^{\Lambda_t^1 + \Lambda_t^2} \sum_{s=t+1}^{T-1} E_Q \left[S_s \left(e^{-\Lambda_{s-1}^1} - e^{-\Lambda_s^1} \right) e^{-\Lambda_{s-1}^2} | \mathcal{F}_t \right]. \end{aligned} \quad (59)$$

Analogously to Proposition 5.3 we obtain that

$$\begin{aligned} E_Q \left[S_s e^{-\Lambda_{s-1}^1 - \Lambda_{s-1}^2 + \Lambda_t^1 + \Lambda_t^2} | \mathcal{F}_t \right] &= e^{a_0 s + \Phi(t,T) + \psi(t+1) \cdot Z_t} \\ E_Q \left[S_s e^{-\Lambda_s^1 - \Lambda_{s-1}^2 + \Lambda_t^1 + \Lambda_t^2} | \mathcal{F}_t \right] &= e^{a_0 s + \Phi'(t,T) + \psi'(t+1) \cdot Z_t} \end{aligned}$$

where we have $\kappa_s = a$, $\kappa_s - 1, \dots, \kappa_{t+1} = -b^1 - b^2$ and $\kappa'_s = a$, $\kappa'_{s-1} = -b^1$, and $\kappa'_{s-1}, \dots, \kappa'_{t+1} = -b^1 - b^2$.

6.3. Longevity. Longevity risk has become increasingly important in insurance valuation, see for example MacMinn et al. (2006), Blake et al. (2014), Blake et al. (2018), and references therein.

From a general viewpoint, longevity comes as a driving systemic factor which affects all insured customers. The generality of our approach allows to incorporate such factors. Moreover, affine processes have already been used to model longevity, see for example Biffis (2005).

For illustrative purposes consider the following example: let $\theta^i : \mathbb{T} \rightarrow \mathbb{R}^+$, $i = 1, \dots, n$ denote n possible mortality curves with initial mixture probability $p^1, \dots, p^n$. For valuation we use the mixture curve

$$\sum_{i=1}^n p^i \theta^i(t), \quad t \in T.$$

If we assume that the curves are ordered, i.e. θ^n is the curve associated with the highest longevity, *longevity risk* describes the possible increase of p^n through time.

Consider a filtration $\mathbb{G}$ and denote by p_t^i the update of the initial probabilities given $\mathcal{G}_t$, which is typically achieved via filtering and Bayes rule, see Schmidt (2016) and Bain and Crisan (2009). Longevity risk arises if chance implies an upward trend in p^n , and affine models are perfectly suited for capturing this: according to Biffis (2005), already random fluctuations in p model longevity risk, while Luciano et al. (2008) specifies an exponentially growing mean-reversion level, which generalizes the Gompertz law, for modeling longevity risk in an affine setting. Both approaches can be embedded in our affine framework.

7. CONCLUSION

This paper deals with the valuation of hybrid insurance products depending on insurance-related quantities like surrender, claim arrival or mortality *and* financial markets. Since interest rates typically have to be taken into account, this applies to most insurance contracts. We fundamentally analyze the valuation of these products by introducing actuarial portfolio strategies combined with financial trading strategies and relate them to the absence of insurance-finance arbitrage (IFA). Moreover, we provide a simple valuation rule which avoids insurance and finance arbitrage. This is illustrated in an affine framework.

Our work can be seen as the first step towards a general study of insurance valuation and future work will need to include non-linear valuation rules, allow for more than one insurance company and should take possible ruin of insurance companies into account.

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